

## List of Content and Research Resources

| Module                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <p><b>Module 1 - Clean burning and wood smoke: Why it matters</b></p> <p><b>Why do people burn wood?</b><br/><a href="https://www.epa.gov/burnwise/wood-smoke-and-your-health">https://www.epa.gov/burnwise/wood-smoke-and-your-health</a></p> <p><b>Other sources of wood smoke other than wood stoves (e.g. slash and burning)</b><br/>Open Burning (from land clearing and forestry) &amp; backyard burning: (<a href="http://breathecleanair.ca/wp-content/uploads/2016/10/Sources-of-wood-smoke-in-the-Comox-Valley.pdf">http://breathecleanair.ca/wp-content/uploads/2016/10/Sources-of-wood-smoke-in-the-Comox-Valley.pdf</a>)</p> <p>Pizza and bagel ovens:<br/><a href="http://ville.montreal.qc.ca/pls/portal/docs/PAGE/ENVIRO_FR/MEDIA/DOCUMENTS/RSQA_BILAN2014_EN.PDF">http://ville.montreal.qc.ca/pls/portal/docs/PAGE/ENVIRO_FR/MEDIA/DOCUMENTS/RSQA_BILAN2014_EN.PDF</a></p> <p>Chimineas: <a href="https://www.lung.ca/news/advocacy-tools/our-position-statements/residential-wood-burning">https://www.lung.ca/news/advocacy-tools/our-position-statements/residential-wood-burning</a><br/><a href="https://en.wikipedia.org/wiki/Chimineas">https://en.wikipedia.org/wiki/Chimineas</a></p> <p>Chimineas: <a href="https://en.wikipedia.org/wiki/Chimineas">https://en.wikipedia.org/wiki/Chimineas</a></p> <p>MoECSS: outdoor fireplaces/chimineas: <a href="https://www2.gov.bc.ca/gov/content/environment/air-land-water/air/air-pollution/smoke-burning/wood-burning-appliances/stoves-inserts">https://www2.gov.bc.ca/gov/content/environment/air-land-water/air/air-pollution/smoke-burning/wood-burning-appliances/stoves-inserts</a></p> <p><b>Poor air quality and the conditions that create it from wood heating</b><br/>Conditions for wood smoke: <a href="http://burningissues.org/car-www/science/temperature-inversion-narr.html">http://burningissues.org/car-www/science/temperature-inversion-narr.html</a><br/><a href="https://www.canada.ca/en/environment-climate-change/services/air-pollution/issues/smog-causes-effects/winter.html">https://www.canada.ca/en/environment-climate-change/services/air-pollution/issues/smog-causes-effects/winter.html</a></p> <p>Health Canada Smog Source: <a href="https://www.canada.ca/en/health-canada/services/air-quality/indoor-air-contaminants/avoid-wood-smoke.html">https://www.canada.ca/en/health-canada/services/air-quality/indoor-air-contaminants/avoid-wood-smoke.html</a></p> <p><b>PM2.5, lung entrance, toxic air pollutants</b><br/>Description of PM2.5 from health Canada: <a href="https://www.canada.ca/en/health-canada/services/publications/healthy-living/guidance-fine-particulate-matter-pm2-5-residential-indoor-air.html">https://www.canada.ca/en/health-canada/services/publications/healthy-living/guidance-fine-particulate-matter-pm2-5-residential-indoor-air.html</a><br/>WHO health effects of particulate matter:<br/><a href="http://www.euro.who.int/_data/assets/pdf_file/0006/189051/Health-effects-of-particulate-matter-final-Eng.pdf">http://www.euro.who.int/_data/assets/pdf_file/0006/189051/Health-effects-of-particulate-matter-final-Eng.pdf</a></p> <p>BC Lung Association:<br/><a href="https://www.lung.ca/news/advocacy-tools/our-position-statements/residential-wood-burning">https://www.lung.ca/news/advocacy-tools/our-position-statements/residential-wood-burning</a></p> <p>Health Canada Source: <a href="https://www.canada.ca/en/health-canada/services/air-quality/indoor-air-contaminants/avoid-wood-smoke.html">https://www.canada.ca/en/health-canada/services/air-quality/indoor-air-contaminants/avoid-wood-smoke.html</a></p> <p>Washington State Department of Ecology document:<br/><a href="https://fortress.wa.gov/ecy/publications/publications/91br023.pdf">https://fortress.wa.gov/ecy/publications/publications/91br023.pdf</a></p> <p><b>Irritation, long-term exposure</b><br/>Health Canada Source: <a href="https://www.canada.ca/en/health-canada/services/air-quality/indoor-air-contaminants/avoid-wood-smoke.html">https://www.canada.ca/en/health-canada/services/air-quality/indoor-air-contaminants/avoid-wood-smoke.html</a></p> |

Canadian Council of Ministers /CCME source  
of the Environment source health impacts:  
[https://www.ccme.ca/files/Resources/air/wood\\_burning/pn\\_1479\\_wood\\_burning\\_code\\_eng.pdf](https://www.ccme.ca/files/Resources/air/wood_burning/pn_1479_wood_burning_code_eng.pdf)

Merriam-Webster Dictionary: <https://www.merriam-webster.com/dictionary/carcinogen>

Washington State Department of Ecology  
document: <https://fortress.wa.gov/ecy/publications/publications/91br023.pdf>

#### **Impact on pre-existing conditions**

Health Canada Source: <https://www.canada.ca/en/health-canada/services/air-quality/indoor-air-contaminants/avoid-wood-smoke.html>

Canadian Council of Ministers of the Environment: Wood Smoke Health Effects:  
[https://www.ccme.ca/files/Resources/air/wood\\_burning/pn\\_1479\\_wood\\_burning\\_code\\_eng.pdf](https://www.ccme.ca/files/Resources/air/wood_burning/pn_1479_wood_burning_code_eng.pdf)

#### **Impact on quality of life / mental well-being**

BC Lung 2016 Presentation Source (Mental Health):  
<https://bc.lung.ca/sites/default/files/media/Mental%20Health.pdf>

Environmental Protection Authority Victoria: <http://www.epa.vic.gov.au/your-environment/air/wood-burning-and-air-quality/why-is-wood-smoke-an-issue>

#### **Energy efficiencies / cost savings**

Energy.gov (US government): <https://energy.gov/energysaver/home-heating-systems/wood-and-pellet-heating>

Energy Nova Scotia: <https://www.efficiencyns.ca/guide/homeowners-guide-to-wood-heating/>

#### **Safety concerns**

Canada Safety Council: <https://canadasafetycouncil.org/woodstove-and-wood-burning-fireplace-safety/>  
<https://www.canada.ca/en/health-canada/services/air-quality/indoor-air-contaminants/keep-carbon-monoxide-out-your-home.html>

<http://www.csia.org/chimneyfires.html>: <http://www.csia.org/chimneyfires.html>

Information to prevent carbon monoxide poisoning: <https://www.canada.ca/en/health-canada/services/air-quality/indoor-air-contaminants/keep-carbon-monoxide-out-your-home.html>

#### **Multiple heating appliances**

<https://www2.gov.bc.ca/gov/content/environment/air-land-water/air/air-pollution/smoke-burning/wood-burning-appliances/stoves-inserts> - Province of British Columbia, Wood Stoves, Inserts, and Outdoor Appliances

#### **Natural gas**

Natural Resources Canada Natural Gas: <http://www.nrcan.gc.ca/energy/natural-gas/5641>

Natural Resources Canada: <http://www.nrcan.gc.ca/energy/products/categories/heating/13740> and <http://www.nrcan.gc.ca/energy/products/categories/heating/gas-fireplaces/16038>

#### **Propane**

Propane appliances: <http://www.housing.yk.ca/eeh.html>

Benefits/challenges heating with natural gas: [http://www.housing.yk.ca/pdf/heating\\_with\\_gas.pdf](http://www.housing.yk.ca/pdf/heating_with_gas.pdf)

### **Heat pumps**

Heat pumps: <https://www.nrcan.gc.ca/energy/publications/efficiency/heating-heat-pump/6827>

### **Electric Heat**

[http://www.housing.yk.ca/pdf/heating\\_with\\_electricity.pdf](http://www.housing.yk.ca/pdf/heating_with_electricity.pdf)

<http://www.housing.yk.ca/eeh.html>

### **Pellet stoves**

<http://www.housing.yk.ca/eeh.html>

<https://www.epa.gov/burnwise/burn-wise-frequently-asked-questions>

## **Module 2 - Firewood Practices**

### **Sourcing firewood**

PGAIR Burn it clean firewood storage:

[http://pgairquality.com/uploads/PGAIR\\_BurnItClean\\_FirewoodStorageBooklet\(WEB\).pdf](http://pgairquality.com/uploads/PGAIR_BurnItClean_FirewoodStorageBooklet(WEB).pdf)

PGAIR Burn It Clean Course: <https://www.pgairquality.com/Programs/master-burner-program>

## **Module 3 - Clean Burning Practices**

### **What to Burn**

PGAIR Burn It Clean Course: <https://www.pgairquality.com/Programs/master-burner-program>

### **What Not to Burn**

Health Canada website: <https://www.canada.ca/en/health-canada/services/air-quality/indoor-air-contaminants/avoid-wood-smoke.html>

PGAIR Burn It Clean Course: <https://www.pgairquality.com/Programs/master-burner-program>

### **Starting a clean fire**

PGAIR Burn It Clean Course: <https://www.pgairquality.com/Programs/master-burner-program>

### **Starting a fire: top-down**

PGAIR Burn It Clean Course: <https://www.pgairquality.com/Programs/master-burner-program>

### **Starting a fire: bottom-up**

PGAIR Burn It Clean Course: <https://www.pgairquality.com/Programs/master-burner-program>

### **Maintaining a clean fire**

PGAIR Burn It Clean Course: <https://www.pgairquality.com/Programs/master-burner-program>

### **Stove and chimney maintenance**

PGAIR Burn It Clean Course: <https://www.pgairquality.com/Programs/master-burner-program>

### **Regular wood stove and chimney inspections**

Canada Mortgage and Housing Chapter 12:

<http://pgairquality.com/uploads/CHMC%20Guide%20To%20Residential%20Wood%20Heating.pdf>

### **Carbon monoxide**

<https://www.canada.ca/en/health-canada/services/air-quality/indoor-air-contaminants/keep-carbon-monoxide-out-your-home.html>

### **Chimney cleaning**

PGAIR Wood burning:

[http://pgairquality.com/uploads/PGAIR\\_BurnItClean\\_WoodburningBooklet\(WEB\).pdf](http://pgairquality.com/uploads/PGAIR_BurnItClean_WoodburningBooklet(WEB).pdf)

### **Wood stoves with catalytic converters**

[https://www.ccme.ca/files/Resources/air/wood\\_burning/pn\\_1479\\_wood\\_burning\\_code\\_eng.pdf](https://www.ccme.ca/files/Resources/air/wood_burning/pn_1479_wood_burning_code_eng.pdf)

## **Module 4 - Wood burning: Provincial and Local Rules and Regulations**

### **Open Burning Smoke Control Regulation**

Guidance Document for Canadian Jurisdictions on Open-Air Burning:

[https://www.ccme.ca/files/Resources/air/wood\\_burning/pn\\_1548\\_CCME%20Guidance%20Document%20on%20Open%20Air%20Burning%20FINAL.pdf](https://www.ccme.ca/files/Resources/air/wood_burning/pn_1548_CCME%20Guidance%20Document%20on%20Open%20Air%20Burning%20FINAL.pdf)

**Open Burning Smoke Control Regulation** <https://www2.gov.bc.ca/gov/content/environment/air-land-water/air/air-pollution/smoke-burning/regulations/openburningregulation>

### **Solid Fuel Burning Domestic Appliance Regulation**

*Environmental Management Act*

Solid Fuel Burning Domestic Appliance Regulation:

[http://www.bclaws.ca/civix/document/id/complete/statreg/218\\_2016](http://www.bclaws.ca/civix/document/id/complete/statreg/218_2016)

### **Factsheet On The Solid Fuel Burning Domestic Appliance Regulation:**

[https://www2.gov.bc.ca/assets/gov/environment/air-land-water/air/reports-pub/sfbdar\\_factsheet.pdf](https://www2.gov.bc.ca/assets/gov/environment/air-land-water/air/reports-pub/sfbdar_factsheet.pdf)

### **Local government and bylaws examples**

British Columbia Ministry of Environment 2015 Inventory of Air Quality Bylaws in British Columbia: Vehicle Idling, Open Burning, and Wood Burning Appliances:

<https://www2.gov.bc.ca/assets/gov/environment/air-land-water/air/reports-pub/bylaws-2015.pdf>

### **Metro Vancouver and air quality regulations**

<http://www.metrovancouver.org/services/Permits-regulations-enforcement/air-quality/Pages/default.aspx>

### **Air quality advisories**

MOECSS: <https://www2.gov.bc.ca/gov/content/environment/air-land-water/air/air-quality/air-advisories>

MOECSS AQHI: <http://www.env.gov.bc.ca/epd/bcairquality/readings/aqhi-table.xml>

PGAIR Burn it Smart Source

### **Throughout Course:**

Canada Mortgage and Housing Corporation's "Guide to residential wood heating":

[http://www.biomassinnovation.ca/pdf/Guide\\_Residential\\_Wood\\_Heating.pdf](http://www.biomassinnovation.ca/pdf/Guide_Residential_Wood_Heating.pdf)

## **List of Video Resources:**

- Wood Smoke and Your Health (French and English) – Created by the Fraser Basin Council with financial support from the Ministry of Environment and Climate Change Strategy, Health Canada, and PGAIR. <https://www.fraserbasin.bc.ca/>

- Burning Clean: It Starts With the Wood (French and English) – Created by the Fraser Basin Council with financial support from the Ministry of Environment and Climate Change Strategy, Health Canada, and PGAIR. <https://www.fraserbasin.bc.ca/>
- Wood Smoke: Tips For a Clean Burn (French and English) – Created by the Fraser Basin Council with financial support from the Ministry of Environment and Climate Change Strategy, Health Canada, and PGAIR. <https://www.fraserbasin.bc.ca/>

### **List of Image and Diagram Sources:**

All Images and diagrams were purchases from Adobe Photo Stock except for the following:

**1) Create Commons: All Creative Commons licensing requirements were followed.**

- Einfache-zeitung (Newspaper)
- Log
- Log-clip-art-log
- GlitchSimplifiedWoodStove

**2) Windows to the Universe:**

Temperature Inversion Diagram

<https://www.windows2universe.org/copyright.html> "Image courtesy of Windows to the Universe."

**3) GRID-Arendal:**

PM2.5 and PM10 Diagram

Diagram Credit: GRD-Adrendal. Available from :[http://old.grida.no/graphicslib/detail/pm25-and-pm10-size\\_0cd](http://old.grida.no/graphicslib/detail/pm25-and-pm10-size_0cd)

**4) Cancer Research UK:**

Diagram showing the lungs including the alveoli CRUK 309

Diagram Credit: Slide 4.10" alv. diagram: By Cancer Research UK - Original email from CRUK, CC BY-SA 4.0, <https://commons.wikimedia.org/w/index.php?curid=34334119>

**5) Firewood Blogspot:**

Chord of Wood

PhotoCredit: [http://4.bp.blogspot.com/\\_fLeiaS6olQ4/TM76xCsblcl/AAAAAAAAAMU/GIzmNdIIISRY/s1600/Firewood%20Cord%20Measurements.png](http://4.bp.blogspot.com/_fLeiaS6olQ4/TM76xCsblcl/AAAAAAAAAMU/GIzmNdIIISRY/s1600/Firewood%20Cord%20Measurements.png)